Geometry : Vectors — problem 12.1.2

A. Engel — Problem-Solving Strategies · 1998

Let A, B, C, D be four space points. Prove the theorem : *If, for all points X in space, $|AX|^2 + |CX|^2 = |BX|^2 + |DX|^2$, then $ABCD$ is a rectangle.*